

Point Calibration Does Not Control Type I Error Across the Composite Null in a Composed Adaptive Bayesian Design

Lu Qian

Zetyra | Evidence in the Wild maggie@zetyra.com

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Abstract

Efficacy thresholds for Bayesian monitoring can be calibrated at a single assumed control rate. We examined whether such point calibration controls Type I error (T1E) across the composite null in a two-arm Phase II design combining historical control borrowing, Bayesian monitoring, sample size re-estimation (SSR), and response-adaptive randomization (RAR). Four isolated numerical checks passed their stated tolerances, although the RAR diagnostic was a stand-alone calculation; none evaluated T1E away from the calibration anchor.

The full-design threshold was calibrated to $\alpha = 0.025$ at $p_0 = 0.25$ with no time trend. Holding that threshold fixed, we varied the common response rate over $p = 0.20$ – 0.45 , with $\delta = 0$, along the equality boundary of the superiority null. Each rejection probability therefore estimates T1E. In the independent grid run, estimated T1E was 0.0236 at the anchor (MCSE 0.0021), 0.0810 at $p = 0.30$, 0.1854 at $p = 0.36$, and 0.2786 at $p = 0.45$, where the curve was still increasing. The two largest rates characterise the shape of the curve rather than plausible control rates for this indication; over $p = 0.27$ – 0.36 estimated T1E ran 0.0380 to 0.1854. Thus, no calendar trend or time-model misspecification was required to produce the control failure.

Disabling SSR and RAR while retaining the full-design threshold did not restore control: the ablation reached 0.2020 at $p = 0.45$, and its contrast with the full design reversed sign across the evaluated range. Because the ablation was not separately calibrated, this contrast does not attribute excess error to individual components. In a separate trend-active stress analysis, results were reported as rejection probabilities rather than T1E for the constant-rate estimand. The pooled $\text{SSR} \times \text{RAR}$ interaction estimate was -0.0077 (95% simulation interval $[-0.0157, +0.0002]$), providing no evidence of a directional interaction.

Point calibration did not provide uniform T1E control. The evaluated grid demonstrates a control violation but does not locate the supremum; operating characteristics should therefore be assessed across prespecified null-generating conditions.

1. Introduction

Computational tools are used to specify priors, simulate operating characteristics and prepare regulatory submissions. In the workflow examined here, the checks applied to such a tool are component-level and numerical: the implemented calculations are examined in isolation. This paper asks what those checks do and do not establish about the error rate of the assembled design. We make no claim about how widespread that workflow is; the question is what follows from it where it is used.

The FDA’s January 2026 draft guidance on the use of Bayesian methodology in clinical trials [U.S. Food and Drug Administration, 2026] is directly relevant. It recommends comprehensive simulation over plausible assumptions, including the background rate of the endpoint and accrual rate; asks sponsors to discuss the handling of prior–data conflict; and, for designs calibrated to a Type I error

target, recommends evaluating Type I error and power. (It is draft guidance containing nonbinding recommendations, and we describe it as such throughout.) These recommendations bear on a specific practice, and one this paper examines directly: an efficacy threshold calibrated at a single assumed control rate, with that single point then treated as evidence of error control.

This paper makes two claims, at two levels. The primary claim is about **robustness**: an efficacy threshold calibrated at one assumed control rate does not deliver Type I error control across the composite null. The secondary claim is about **composition**: at the frozen full-design threshold, the combined SSR/RAR contrast changes sign across the evaluated null rates, but the factorial interaction estimates are too imprecise to support a stable interaction claim in either direction.

Both claims are demonstrated by simulation over a grid of common baseline rates. We are explicit about what such a grid can and cannot establish: it exhibits points at which control fails, but it does not locate or estimate the supremum of the Type I error over the null.

The case study was developed with Zetyra; the released R/Quarto implementation is the reproducibility artifact for this paper. The platform supports the argument; it is not the argument itself.

2. Background

2.1 What Component-Level Checks Establish

“Validation” of clinical trial design software is not a term with a single agreed meaning. Where it amounts to documenting that design outputs were computed with a named package, with cross-checks against published values, it does not establish maximum numerical deviation across the range of design parameters, does not guarantee reproducibility under stochastic methods, and does not account for the behaviour of multiple adaptive components operating jointly. The FDA January 2026 draft guidance addresses this directly: it recommends that operating characteristics be evaluated by comprehensive simulation over plausible assumptions, discusses prior–data conflict explicitly, and recommends scenarios spanning plausible values of design inputs—including the background rate of the endpoint—documented reproducibly. These are nonbinding recommendations in a draft document.

The case study was developed with Zetyra. The released R/Quarto source and seeds provide the code needed to rerun the primary Rule B analysis (see Acknowledgements); exact cross-environment numerical identity is not claimed. Results were produced under R 4.6.1 with RBesT 1.10-0, rstan 2.32.7 and posterior 1.7.0; the full `sessionInfo()` is written to the release on each render. The versions matter because the historical-control prior is fitted through RBesT and every operating characteristic reported here conditions on that fitted mixture, so the render asserts the mixture against a frozen full-precision reference and stops if it moves. The public validation repository documents the broader software checks.

2.2 Composed Adaptive Designs and Marginal Calibration

Definition. A *composed adaptive design* is a design in which two or more adaptive mechanisms act at different times on overlapping information from the evolving trial, before the final analysis. They need not share a schedule or a view of the data, and in the design studied here they do not. RAR updates the allocation probability patient by patient from arm-labelled outcomes, after a ten-patient burn-in. The SSR rule fires once, at $n = 45$, on pooled counts with the arm labels

withheld. Bayesian monitoring evaluates arm-labelled data at the scheduled looks $\{30, 45, 60, 90\}$. What they share is the accruing trial itself: each rule alters the data the others subsequently see.

We do not formalise this as a set-theoretic condition on information sets. Writing $\mathcal{F}_i \cap \mathcal{F}_j \neq \emptyset$ would not distinguish anything: for sigma-fields the intersection always contains $\emptyset$ and Ω , so the condition holds trivially for any pair. Nor is the converse available—informational independence does not by itself guarantee that error control survives adaptation.

The relevant observation is weaker and purely negative. Let $\phi(X)$ denote the final rejection rule (Bayesian monitoring), and let $A_1(X_t)$, $A_2(X_t)$ denote the adaptation rules—SSR and RAR—which alter sample size or allocation based on interim data X_t but do not themselves reject H_0 . Each is checked in isolation (the blinded nuisance-parameter SSR satisfies its pooled-input, algebra and bound checks; RAR tracks $P(\theta_e > \theta_c)$ within tolerance). The composed decision rule is $\phi_{\text{comp}}(X) = \phi(X; A_1(X_t), A_2(X_t))$: the final rejection indicator evaluated on data whose sample size and allocation were shaped by the adaptation rules. The workflow examined here calibrates that composed rule at a single point, choosing γ^* so that $P_{p_0, \delta=0}(\phi_{\text{comp}}(X) = 1) \approx \alpha$ at one anchor rate with no trend. That targets the error rate at that point but places no bound on $P_{p, \delta}(\phi_{\text{comp}}(X) = 1)$ for other (p, δ) in the null. The component numerical checks add no such bound either, since none of them evaluates an error rate.

Point calibration alone does not establish range control. Checking components in isolation, and calibrating the composed rule at a single null point, places no uniform bound on the error rate of the composed rule over the null. The claim is about a bound, not about every point: continuity in p may constrain the error rate in a neighbourhood of the anchor, and the grid below shows it does—0.0236 at $p = 0.25$ against 0.0380 at $p = 0.27$. Nothing in the workflow constrains it away from one. This is a statement about what those checks fail to establish, not a condition under which inflation must occur. Whether it occurs, in which direction and by how much are empirical questions, which is why the remainder of the paper is a simulation study. Figure 1 illustrates the structure.

3. Simulation Study

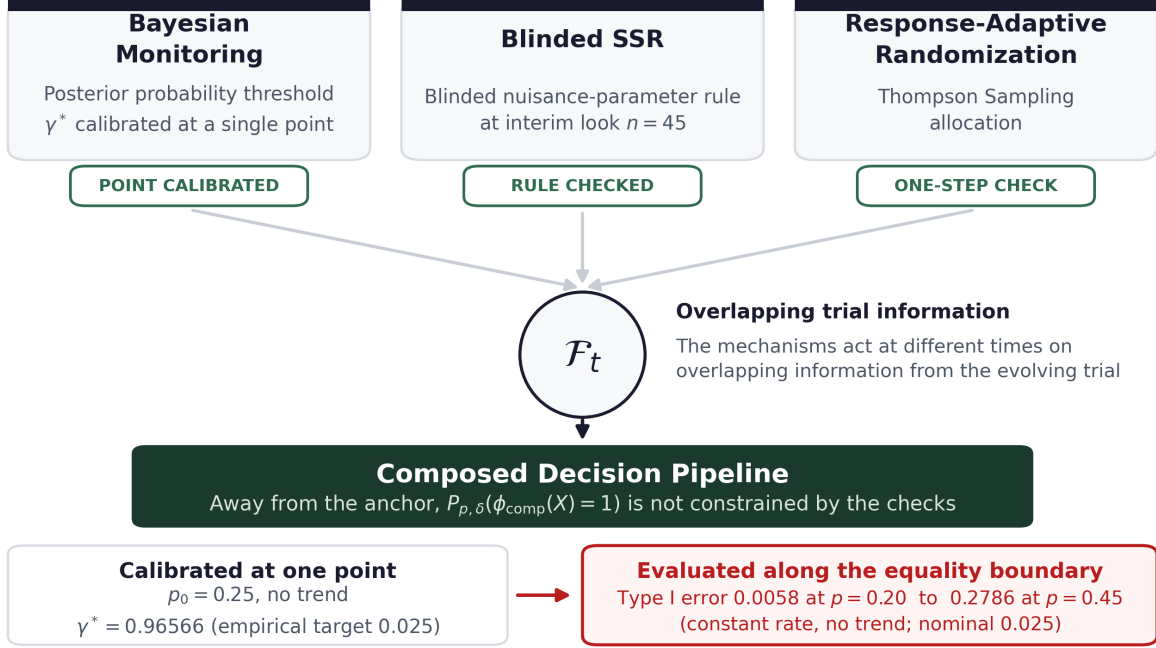
3.1 Design Specification

We simulate a two-arm Phase II oncology design with MAP borrowing on the control arm [Weber et al., 2021], interim looks at $n = 30, 45, 60$ and a final at $n \geq 90$ (post-SSR). RBesT fits the historical cohorts ($n = 50, 40, 60$; 38/150 responses) as a robustified mixture with components

$$0.5887 \text{Beta}(18.403, 53.273), \\ 0.2113 \text{Beta}(1.956, 4.703), \quad \text{and} \quad 0.2000 \text{Beta}(1, 1).$$

The experimental-arm monitoring prior is $\text{Beta}(0.5, 0.5)$. RAR maintains separate $\text{Beta}(0.5, 0.5)$ posteriors for the two arms, alternates allocation for a 10-patient burn-in, and thereafter assigns the experimental arm when an independent draw from its posterior exceeds the corresponding control-arm draw. At 50% enrollment, Rule B uses only the pooled response count to estimate the overall event-rate nuisance parameter, while the planned treatment effect $\Delta = 0.45 - 0.25 = 0.20$ remains fixed. Under a prespecified working 1:1 allocation, it preserves the fixed-sample Wald power implied by the original $N = 90$ design (0.5296), permits increases only, and applies the protocol cap $N = 150$. The working rates are

$$\hat{p}_c = \text{clamp}_{[10^{-6}, 1-\Delta-10^{-6}]}(\hat{p}_{\text{pool}} - \Delta/2), \quad \hat{p}_e = \hat{p}_c + \Delta.$$



Arrows: data flow from the accruing trial. Badges state only the check shown; none is an operating-characteristic evaluation across the null.

Figure 1: Composed adaptive design: Bayesian monitoring, blinded nuisance-parameter SSR and RAR act at different times on overlapping information from the evolving trial—patient by patient, once at $n = 45$ on blinded pooled counts, and at the scheduled looks respectively. Component checks and point calibration at the anchor do not bound Type I error elsewhere in the null. The grid evaluated here is the equality boundary $p_e = p_c$, not the full composite null.

The recalculated per-arm size is

$$n_{\text{arm}} = \frac{(z_{1-\alpha} + z_{\text{power}})^2 [\hat{p}_c(1 - \hat{p}_c) + \hat{p}_e(1 - \hat{p}_e)]}{\Delta^2},$$

rounded up and doubled. This pooled-data construction follows the blinded binary internal-pilot principle of Friede and Kieser [2004]. It is a conventional-style working nuisance rule, not an exact power calculation for the Thompson-sampling allocation or Bayesian final decision. At the planned $n = 45$ look, its complete count mapping spans only $N = 90$ – 100 ; the exact expected N is 90.44 at $p = 0.25$, 91.57 at $p = 0.30$, and 94.12 at $p = 0.36$ under the active trend. Thompson Sampling RAR is accompanied by a stand-alone one-step check against analytical $P(\theta_e > \theta_c)$; that diagnostic does not invoke the production allocation path in the trial simulator.

The threshold is calibrated at the design anchor $p_0 = 0.25$, distinct from the dominant-component mean 0.2568, robustified-mixture mean 0.3132 and historical pooled rate 0.2533. We therefore describe $p > 0.25$ only as departure from the calibration anchor, not from the prior mean.

For enrollment position i , the response probabilities are $p_e(i) = p_e + \delta(i - 1)/89$ and $p_c(i) = p_c + \delta(i - 1)/89$, with the evaluated values inside $[0, 1]$. The superiority null is $H_0 : p_e \leq p_c$. The constant-rate sweep evaluates its equality boundary, setting $p_e = p_c = p$ and $\delta = 0$. A violation at any such point is sufficient to disprove uniform control, but the sweep does not establish that the boundary is least favorable or characterize the interior $p_e < p_c$. The trend-active stress sweep also sets the baseline rates $p_e = p_c = p$ and uses $\delta = 0.05$, so $p_e(i) = p_c(i)$ at every position while both

drift; the analysis remains time-constant.

The simulation completes enrollment to the total selected by Rule B and then evaluates the maximum posterior probability at total enrollments 30, 45, 60, 90, plus the realised final sample size when it exceeds 90. Because allocation, outcomes and SSR do not depend on the efficacy boundary, thresholding this maximum preserves the efficacy-only reject/no-reject indicator that prospective monitoring with immediate outcomes would produce. It does not simulate operational early stopping and therefore does not estimate efficacy stopping times, early-stopped sample sizes, or allocations after first boundary crossing.

Under H_0 both arms share baseline rate p and drift together, so there is no contemporaneous treatment effect; but the analysis is time-constant while the data drift, so these quantities are kept outside the paper’s constant-rate Type I error estimand and reported as rejection probabilities. Within the wider $p = 0.20\text{--}0.45$ trend grid, we highlight $p = 0.27$ and $p = 0.30$ with linear trend $\delta = 0.05$ over the planned enrollment horizon. These are sensitivity scenarios, not an asserted empirical range; Tang et al. [2010] and Viele et al. [2014] provide clinical and historical-borrowing context.

All primary Rule B estimates carry binomial MCSE $\sqrt{\hat{p}(1 - \hat{p})/N_{\text{sim}}}$; cell-specific MCSEs and 95% simulation intervals are reported here or in the simulation supplement. Downstream MCSEs and intervals condition on the selected γ^* and do not propagate its calibration-sample uncertainty; the separate bootstrap below describes threshold-selection variability only.

Note on calibration: The efficacy threshold is obtained by grid-free binary search, giving $\gamma^* = 0.96566$ with in-sample T1E = 0.0248 ($N = 20,000$) at the design anchor $p_0 = 0.25$ with no trend. No MCSE is quoted for that value: γ^* is chosen by bisection on this same sample, so the achieved level is near-deterministic by construction. The calibration bootstrap gives γ^* standard deviation 0.00106, 95% interval [0.96303, 0.96759] over 500 resamples. On an independent fresh-seed sample ($N = 5,000$), T1E was 0.0224 (MCSE 0.0021; 95% simulation interval [0.0183, 0.0265]), compatible with the nominal target. Binary search still targets an empirical sample rather than the population error rate, so neither estimate is exact calibration.

3.2 Component-Level Numerical Checks

Four isolated calculations were checked. MAP prior: maximum deviation from the RBest mixture = 1.78×10^{-15} (floating-point precision). Both sides of that comparison are computed from the same fitted mixture, so it is an internal consistency check on the fitted object, not an external benchmark. Posterior probabilities use deterministic 400-node Gauss–Legendre quadrature. A targeted comparison of 12 representative reachable states nearest γ^* against adaptive integration found maximum deviation 3.83×10^{-12} and no decision flips; this was a targeted boundary check, not an exhaustive enumeration. Rule B SSR: the calculation reproduces $N = 90$ at the exact planning nuisance rate, uses only the pooled interim count, is increase-only, returns even sample sizes, and stays within the prespecified bounds; its full $n = 45$ input support was exhaustively checked. The pooled nuisance construction is compared with the framework of Friede and Kieser [2004], but the embedded working Wald rule is not claimed to reproduce the operating characteristics of their final test. RAR one-step allocation: a stand-alone diagnostic has maximum deviation from analytical $P(\theta_e > \theta_c)$ of 0.0036 (Monte Carlo error at $n = 100,000$). It reimplements one allocation draw and does not exercise the production allocation path inside the trial simulator. All four checks pass their stated tolerances; none evaluates Type I error away from the anchor.

3.3 Constant-Rate Equality Boundary of the Composite Null

We first hold $\delta = 0$, so the response rate is constant over calendar time and identical in both arms at every point of the grid. Every point is then a genuine null for the analysis actually performed, and every rejection is a Type I error in the conventional sense. This isolates the robustness question from any question about model misspecification.

p	SSR + RAR disabled	All mechanisms	MCSE (full)	Fixed- γ^* difference
0.20	0.0298	0.0058	0.0011	−0.0240
0.22	0.0356	0.0090	0.0013	−0.0266
0.25	0.0534	0.0236	0.0021	−0.0298
0.27	0.0780	0.0380	0.0027	−0.0400
0.30	0.0936	0.0810	0.0039	−0.0126
0.33	0.1330	0.1314	0.0048	−0.0016
0.36	0.1562	0.1854	0.0055	+0.0292
0.40	0.1778	0.2358	0.0060	+0.0580
0.45	0.2020	0.2786	0.0063	+0.0766

At the calibration anchor $p = 0.25$ the full design attains 0.0236 (MCSE 0.0021), consistent with the nominal 0.025. The estimated Type I error increased at every successive evaluated grid point: 0.0380 at $p = 0.27$, 0.0810 at $p = 0.30$, 0.1854 at $p = 0.36$ and 0.2786 at $p = 0.45$. **Calibration at a point does not deliver control over the composite null, and no calendar trend or time-model misspecification is needed to show it.**

The two largest rates characterise the shape of the curve; we do not assert that they are plausible control rates for this indication. We note also that $p = 0.45$ is a *null* scenario in which both arms sit at 0.45, so there is no treatment effect; the value is numerically equal to the experimental-arm rate assumed under the planning alternative (against a 0.25 control), but the coincidence is arithmetic only and that row is not an alternative-hypothesis evaluation. Over the lower evaluated range $p = 0.27$ – 0.36 —departures of two to eleven percentage points from the anchor—Type I error runs from 0.0380 to 0.1854, up to 7.4 times nominal.

3.4 Trend-Active Stress Analysis

We next activate a shared calendar trend in both arms while retaining the time-constant analysis. Contemporaneously there is still no treatment effect, but the analysis is misspecified for the time-varying data-generating process. We therefore report both columns as rejection probabilities rather than Type I errors for the constant marginal-rate null. This caution applies even when RAR is disabled. The implemented rule, $\text{arm} = (i - 1) \bmod 2$, gives deterministic strict alternation: control occupies odd enrollment positions and experimental treatment occupies even positions. By construction, this balances counts at even looks and tightly interleaves enrollment time, but it is not exact time balance under the coded patient-level trend. At every even look, experimental patients are one position later within each pair, so their average generating rate exceeds control by $\delta/(N_{\text{base}} - 1) = 0.05/89 \approx 0.00056$; at the odd $n = 45$ look, arm counts differ by one. With RAR active the allocation additionally depends on accumulating data. This study does not identify a time-adjusted estimand or measure either allocation–time mechanism.

p	SSR + RAR disabled	All mechanisms	Fixed- γ^* difference
0.20	0.0342	0.0092	-0.0250
0.22	0.0474	0.0208	-0.0266
0.25	0.0742	0.0484	-0.0258
0.27	0.0900	0.0754	-0.0146
0.30	0.1180	0.1206	+0.0026
0.33	0.1398	0.1670	+0.0272
0.36	0.1644	0.2286	+0.0642
0.40	0.1894	0.2798	+0.0904
0.45	0.1970	0.2874	+0.0904

Two readings follow. First, **at the full-design threshold, disabling SSR and RAR does not return the rejection probability to the nominal level**: with both switched off, it still rises from 0.0342 at $p = 0.20$ to 0.1970 at $p = 0.45$. That column is the same fixed-threshold ablation used throughout (Section 3.5)—it carries γ^* , which was calibrated with SSR and RAR active. At $p = 0.25$ on this trend-active slice it is 0.0742; neither that row nor the full-design row is the no-trend calibration scenario. Read it as “the design with the adaptive components switched off, threshold unchanged”, not as a calibrated monitoring-only design. Second, **at fixed γ^* , enabling SSR and RAR lowers the rejection probability at low p and raises it at high p** : the difference runs from -0.0250 at $p = 0.20$ to +0.0904 at $p = 0.45$, reversing sign across the range.

The grid establishes a direction change, not its crossing point: the contrast at $p = 0.30$ is +0.0026 against a Monte Carlo standard error of 0.0065, and the neighbouring rates are 0.27 and 0.33. It also shows elevated rejection probabilities across the grid but does not locate their supremum. The largest *estimated* rejection probability on this grid is 0.2874; that is a Monte Carlo estimate carrying an MCSE of 0.0064, not a deterministic bound, and a one-sided 95% simulation limit places that rate’s rejection probability at or above 0.2769.

Comparing this grid with the constant-rate grid of Section 3.3 at matched baseline rates, the trend’s contribution is *not* monotone in the departure. The trend-active point estimate is higher throughout: the difference rises from +0.0034 at $p = 0.20$ to +0.0396 at $p = 0.30$, dips to +0.0356 at $p = 0.33$, reaches +0.0440 at $p = 0.40$, and falls to +0.0088 at $p = 0.45$. The first gap is borderline (2.0 Monte Carlo standard errors) and the last is indistinguishable from zero (1.0 standard error), so no uniform separation is claimed. The fall from $p = 0.40$ to $p = 0.45$ is -0.0352, about 2.8 standard errors, and rests on one point at the grid boundary. We report it as an observation about that point and offer no mechanism. The more aggressive Rule A stress test is summarized in Appendix A.

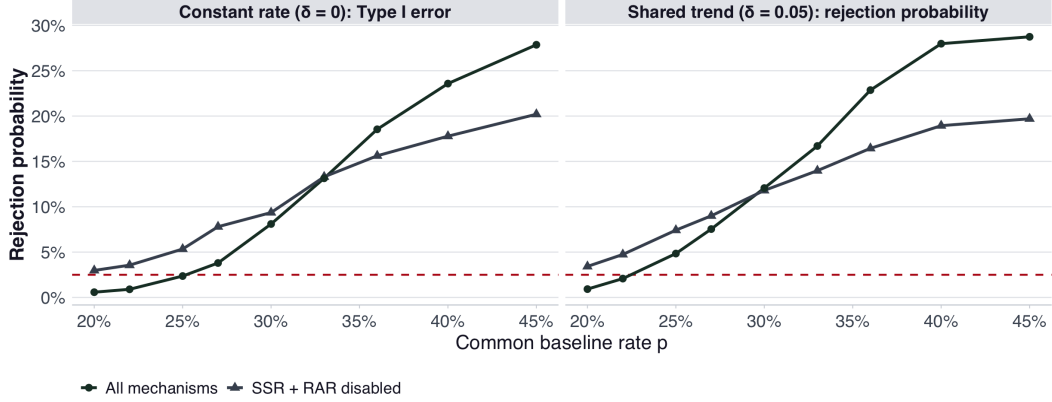


Figure 2: Rejection probability across the common-rate grid. **Left panel:** the constant-rate equality boundary of the superiority null ($\delta = 0$), where the time-constant likelihood matches the data-generating process and every rejection is a Type I error. **Right panel:** the trend-active stress scenario ($\delta = 0.05$), where both arms drift together but the analysis remains time-constant; those rejections are kept outside the constant-rate Type I error estimand and are reported as rejection probabilities. In both panels, grey holds the full-design threshold with SSR and RAR disabled and dark green shows all mechanisms; both are fixed-threshold comparisons, not separately calibrated designs. Dashed red line: nominal $\alpha = 0.025$.

3.5 Fixed-Threshold Factorial Ablation: SSR by RAR

Four designs ($\text{SSR} \times \text{RAR}$), all at $p = 0.30$ with the trend active:

Condition	Rejection probability	MCSE
SSR + RAR disabled	0.1120	0.0045
+ SSR only	0.1248	0.0047
+ RAR only	0.1236	0.0047
+ SSR and RAR	0.1124	0.0045

These are ablations at a fixed threshold, not a decomposition. γ^* was calibrated with SSR and RAR active; disabling components while retaining it does not yield separately calibrated designs or attributable shares. The baseline cell is therefore not “the excess caused by monitoring”, although it does show that disabling both components leaves the rejection probability far above the nominal level. These cells are trend active, so the quantity elevated here is a rejection probability, not a Type I error.

The SSR-enabled cells can accrue an additional efficacy analysis. The monitoring schedule is $\{30, 45, 60, 90\}$ together with the realised final sample size, so a trial that SSR extends beyond $N = 90$ is analysed five times while a fixed- N trial is analysed four. Because the test statistic is a maximum over looks, the SSR-enabled cells receive one further opportunity to cross the threshold in the $\Pr(N > 90)$ fraction of replicates, which is 0.3157 at $p = 0.30$. This is correct trial conduct – an extended trial does analyse at its realised final N – but it means every full-design-versus-ablation contrast reported here combines the adaptive mechanisms with a difference in multiplicity, and no contrast should be read as isolating the mechanisms alone.

The descriptive simple-contrast point estimates are $\Delta_{\text{SSR}|\text{RAR}=0} = +0.0128$ and $\Delta_{\text{RAR}|\text{SSR}=0} =$

+0.0116; the combined baseline-referenced contrast is +0.0004, with approximate Monte Carlo standard errors 0.0063–0.0065. The higher-precision replicate ($N = 20,000$ per cell) estimates the same combined contrast at +0.0092, and pooling the two gives +0.0074 (SE 0.0029, 95% SI [+0.0018, +0.0130]), which excludes zero. We still draw no directional component conclusion, and the reason is structural rather than a matter of precision: γ^* was calibrated once with every mechanism active and each ablation retains it, so these cells are not separately calibrated designs and no contrast between them partitions the excess. More replicates would sharpen the contrast without making it attributable.

The interaction was estimated on two disjoint seed blocks using the same implementation. The first run ($N = 5,000$ per cell) gave -0.0240 (propagated MCSE 0.0091, 95% SI $[-0.0419, -0.0061]$), which excludes zero. A higher-precision replicate on a disjoint seed block ($N = 20,000$ per cell) gave -0.0037 (MCSE 0.0046; 95% SI $[-0.0126, +0.0052]$), which does not. The two are 2.0 standard errors apart (heterogeneity $Q = 3.96$ on 1 df, $p = 0.047$) – near the edge of what independent sampling produces unaided, and not on its own evidence of a defect. Their inverse-variance pooled estimate is $\Delta_{\text{interaction}} = -0.0077$ (SE 0.0041; 95% SI $[-0.0157, +0.0002]$).

We report the pooled value and draw no directional conclusion. The lower-precision run’s interval excluding zero is not reproduced by the better-powered replicate, and we do not interpret it on its own. Joint evaluation remains necessary because an interaction is possible, but its sign and magnitude should not be inferred from these runs.

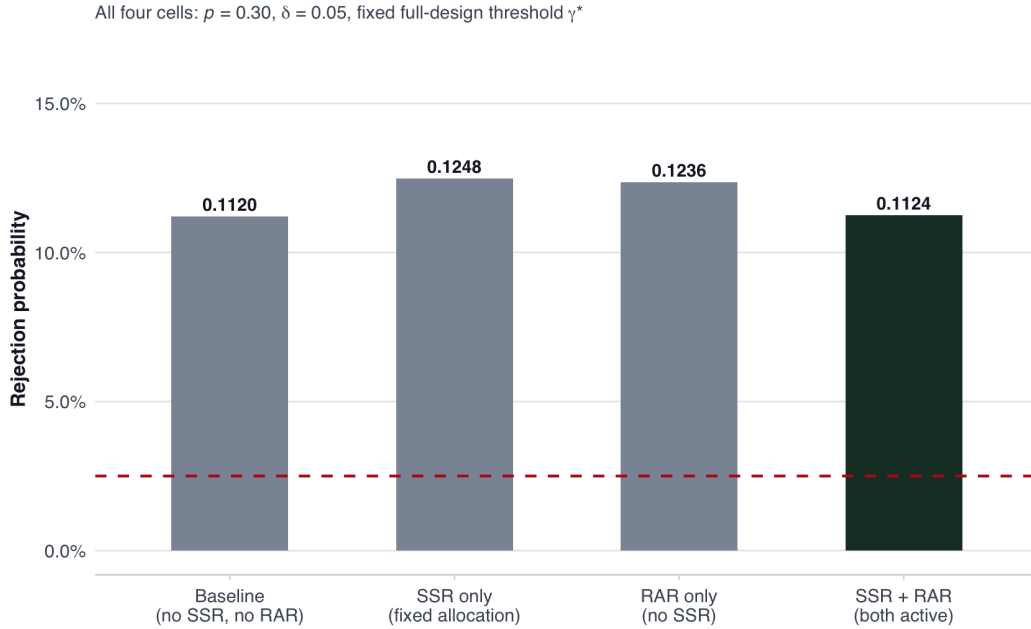


Figure 3: Fixed-threshold Rule B factorial ablation of SSR by RAR at $p = 0.30$ with the shared trend active. Simple contrasts $\Delta_{\text{SSR}|\text{RAR}=0} = +0.0128$ and $\Delta_{\text{RAR}|\text{SSR}=0} = +0.0116$. The interaction is estimated on two disjoint seed blocks; the inverse-variance pooled estimate is -0.0077 (95% SI $[-0.0157, +0.0002]$), which includes zero. Bar labels are cell estimates; the interaction is not annotated in the panel because it is a four-cell contrast rather than a property of any one bar. Dashed red line = nominal $\alpha = 0.025$.

3.6 Baseline-Rate Departure by Time Trend

A 2×2 factor crossing evaluates departure and trend under the primary design; both factors feed monitoring, SSR and RAR, so it does not isolate those channels.

Condition	Rejection probability	vs. anchor run
Anchor, no trend (independent run)	0.0232	—
Departure only	0.0904	+0.0672
Trend only	0.0474	+0.0242
Both	0.1140	+0.0908
Additive prediction	0.1146	
Interaction term	−0.0006	

The departure-only simple contrast is +0.0672, and the trend-only simple contrast is +0.0242. Because both factors feed monitoring, SSR and RAR, neither contrast is attributable to a single channel. Unlike the fixed-threshold ablations, all four cells use the full design and the same independently simulated anchor/no-trend reference condition. The interaction is −0.0006 (MCSE 0.0071; 95% SI $[-0.0145, +0.0133]$). Its point estimate is consistent with additivity at these factor levels, while the interval allows modest departures in either direction. It is a risk-difference interaction at these factor levels and includes the additional trend exposure from enrollment beyond $N = 90$; Rule B limits that extension to 100 at the planned SSR look.

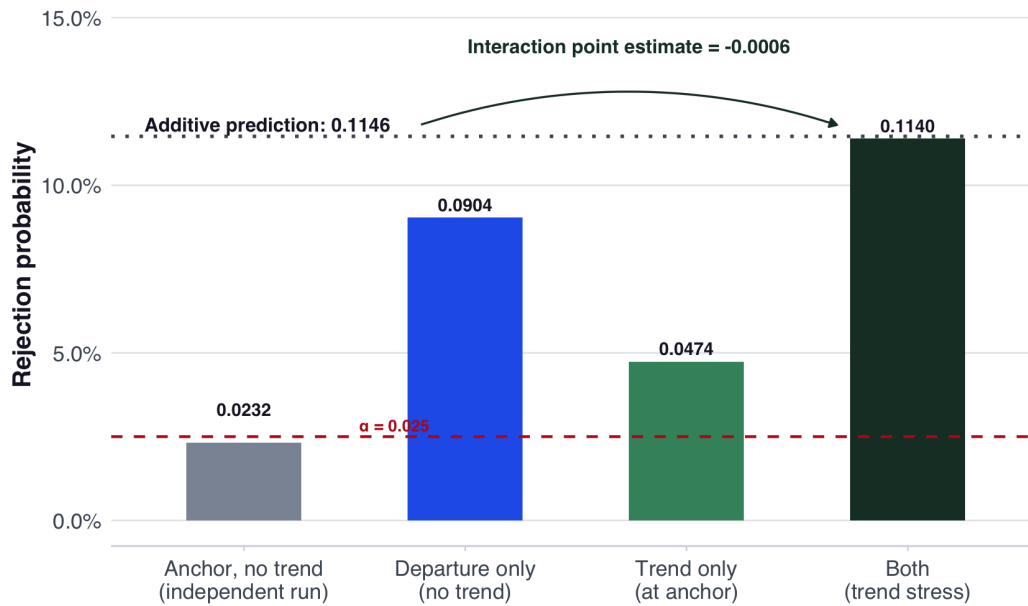


Figure 4: Departure-by-trend crossing under the full design. The interaction is −0.0006 (95% SI $[-0.0145, +0.0133]$); dashed red line denotes nominal $\alpha = 0.025$.

3.7 Sensitivity Analyses

The following sweeps use the five-percentage-point departure ($p = 0.30$) to characterize mechanism sensitivity across a wider range.

- **Departure severity**, at constant trend ($p \in \{0.25, 0.27, 0.30\}$, all with $\delta = 0.05$): the rejection probability increases monotonically, $0.0484 \rightarrow 0.0754 \rightarrow 0.1206$, with no direction reversal.
- **Vague-component weight** (0–90%): larger weight on the flat robustify component means less borrowing; the rejection probability declines from 0.1488 to 0.0624.
- **Time trend** δ (0–15%): the rejection probability increases from 0.0812 at $\delta = 0$ —which is a Type I error, the trend being absent—to 0.2220 at $\delta = 0.15$.
- **SSR trigger timing**: moving the primary look from $n = 45$ to $n = 30$ changed the rejection probability from 0.1161 to 0.1244; a separate sweep over $n_{\text{SSR}} \in \{20, 30, 40, 45, 55, 60\}$ ranged from 0.1126 to 0.1234. Timing also changes the realised final- N distribution, so these are not isolated timing effects.

4. Discussion

4.1 Regulatory Implications and Recommendations

An efficacy threshold calibrated at one assumed control rate need not control Type I error elsewhere in the composite null. The FDA January 2026 draft guidance recommends comprehensive simulation over plausible assumptions and asks sponsors to discuss the handling of prior–data conflict. We go further, on the strength of the results above: operating-characteristic evaluation should span a prespecified set of null-generating conditions, and the handling of prior–data conflict should be prespecified rather than discussed post hoc. That extension is ours, not the guidance’s.

For designs of the kind studied here—binary endpoint, MAP borrowing on the control arm, and a threshold calibrated at a single assumed control rate—sponsors should (1) evaluate a grid of plausible control rates rather than one anchor; (2) label whether component ablations are separately calibrated, since a fixed-threshold ablation tests restoration of control rather than attribution; and (3) cross departure and time trend rather than rely only on separate one-factor analyses.

4.2 Mitigation Strategies

Potential safeguards include more robust or dynamic borrowing, pre-specified prior–data disagreement diagnostics, pipeline-level recalibration of γ^* over the null grid, and an explicit maximum acceptable T1E over a stated robustness range. Each changes the composed pipeline and must itself be evaluated at pipeline level; no component-level argument guarantees restored control.

4.3 Limitations and Future Work

This paper demonstrates a control failure by simulation in one specific design. We did not evaluate other borrowing methods, endpoints, sample sizes or α levels, and the recommendations above are offered as reasonable practice rather than as results. Several further limitations bound what is established here.

The supremum is not estimated. The grid exhibits points at which control fails. The largest *estimated* Type I error on the constant-rate grid is 0.2786, which is a Monte Carlo estimate with an MCSE of 0.0063 and may lie either side of the true value at that rate; a one-sided 95% simulation limit places the Type I error at $p = 0.45$ at or above 0.2682. Because the supremum is at least the value at every evaluated point, the same pointwise limit is also a lower confidence bound for the supremum—conditional on the selected γ^* . Its 95% coverage is with respect to Monte Carlo error at that rate only; it does not propagate the calibration-sample uncertainty in the threshold (bootstrap SD 0.00106), and is therefore not an unconditional statement. Neither figure locates the worst

case, and the curve has not flattened at the grid boundary. The final three increments are +0.0540, +0.0504 and +0.0428, but they span unequal steps in p (0.03, 0.04, 0.05); per percentage point of p the increase is 0.0180, 0.0126 and 0.0086, so the series is decelerating while still rising. The supremum over the composite null is therefore not located, and may lie above any value evaluated here. Locating the supremum, estimating it sharply, or constructing an upper confidence bound would require a search beyond this finite grid.

Composition is not established as the cause. The title and Section 2.2 describe the setting: a design in which several adaptive mechanisms act on overlapping information. The study does not show that composition produced the failure. It cannot: γ^* is calibrated once with all mechanisms active and every ablation retains that threshold, so the ablation rows answer whether switching a mechanism off restores control, not whether composition caused the loss. Answering the causal question would require separately calibrated designs at each configuration, which we did not run. Relatedly, we evaluated no zero-borrowing condition—the vague-component sweep reaches 90% weight, not 100%—so the contribution of MAP borrowing cannot be separated from that of monitoring and adaptation.

No power evaluation is reported for the composed design. The only power figure here is the 0.5296 fixed-sample Wald reference implied by $N = 90$. The safeguards in Section 4.2 would each change the operating characteristics, and recalibrating γ^* over a null grid would raise the threshold and cost power; we did not quantify that cost. None of the safeguards was simulated. They are hypotheses about what would restore control, not results.

Only the equality boundary is evaluated. The constant-rate grid covers $p_e = p_c = p$, a one-dimensional boundary slice of $H_0 : p_e \leq p_c$. Failure at any one of those points is enough to refute uniform control, but the simulations do not establish the least-favorable configuration or characterize the interior $p_e < p_c$.

Conflict is not quantified. We vary the calibration-anchor departure, not a prior-predictive conflict measure. Extending prior-data diagnostics such as Matthews and Qian [2026], Qian and Matthews [2026] from Normal-mean settings to this binary MAP-mixture design would require a new statistic and calibration.

Rule B is a working nuisance calculation, not exact design power. Its 1:1 Wald model differs from the RAR allocation and Bayesian final decision; the 0.5296 reference power and 90–100 mapping follow from the original $N = 90$ design. Results are conditional on this prespecified rule. Appendix A summarizes the more aggressive Rule A stress test.

No sufficient condition is proved. Shared information neither guarantees nor precludes error control; a conditional-error or design-specific argument would be required.

Calibration is Monte Carlo. Binary search targets the empirical calibration sample; a fresh-seed validation was compatible with nominal, but neither finite simulation establishes the population error rate exactly.

Numerical integration is implementation-specific. Posterior probabilities use the stated 400-node Gauss–Legendre rule. The 12-state near-boundary comparison found no decision flips, but it was not an exhaustive audit of all reachable states and does not establish invariance to a different integration rule. The reported operating characteristics are conditional on this implementation.

The implementation is efficacy-only. Retrospective evaluation of the maximum posterior probability preserves the reject/no-reject indicator because the boundary does not drive allocation, outcomes or

SSR, but it does not reproduce operational early stopping or estimate stopping-time and post-stop quantities.

Outcome timing and accrual are simplified. Outcomes are available immediately for per-patient RAR updates, and enrollment index is used as a proxy for calendar time. Delayed outcome availability and stochastic accrual are not modeled.

Component ablations are not separately calibrated. They retain the full-design γ^* and answer whether disabling a component restores control, not how much excess it caused. Attribution would require calibrating each design separately.

5. Conclusion

Point calibration did not provide uniform Type I error control across the composite null. At the full-design threshold, disabling SSR and RAR did not restore it on the constant-rate grid. The fixed- γ^* contrast reversed sign across the evaluated null rates, but the Rule B factorial and departure-by-trend interaction intervals both included zero; the simulations do not support a stable interaction claim in either direction.

The practical consequence is a reporting standard rather than a new method. Operating characteristics should be evaluated over a prespecified grid of null-generating conditions. Fixed-threshold ablations can show whether disabling a component restores control, but attribution requires separately calibrating each design. The simulation code is publicly available for independent verification and extension; the Zetyra platform is a commercial SaaS product.

Calibrate at a point. Validate over the null.

Acknowledgements

Simulation source, manuscript source, figures and reproducibility instructions: github.com/evidenceinthewild/Composed-Adaptive-Trial-Validation. Rule A is summarized in Appendix A as a method-development stress test, not as the primary analysis.

A. Original Rule A Pooled Conditional-Power Stress Test

The original Rule A treated the pooled interim rate as a one-sample signal against $p_0 = 0.25$ and increased total sample size to target 80% conditional power, capped at $N = 150$. A common upward nuisance shift under the two-arm null could therefore drive expansion, so it was replaced before publication by Rule B. With $\gamma^* = 0.96596$, Rule A reached a higher rejection probability at $p = 0.36$ under the trend-active null grid than Rule B's 0.2286 at the same rate. We give no figure for it: the Rule A run is not part of the primary release, and its simulation budget and Monte Carlo standard error are not reported here, so a four-decimal value would carry an authority nothing released can support. The comparison is retained only to record why Rule A was replaced.

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